

PS-NAV — Probability & Statistics (MT-335) — Master Index

Weeks 1–15 · 17 Aug – 29 Nov 2026 · **Midterm 35** / **Final 45** / **Sessional 20** · load 3.0 · difficulty 3/5

Open this file first. It is the map: where every topic lives, every line offset, every resource. **Three-file pack** (all in `terms/2026-fall/PS/`): | **File** | **Holds** | `---|---` | `PS-NAV.md` | This map — glance tables, master offsets, resources index | | `PS-BEFORE-MID.md` | **Weeks 1–8** (midterm) — full per-week content, modules 1–5 | | `PS-AFTER-MID.md` | **Weeks 9–15** (final) — full per-week content, modules 6–8 |

Where to look (fast answers)

I want...	Open
A topic with its defs/formulas/numericals	that week's block in BEFORE-MID / AFTER-MID
Any week's exact line offsets	Master offset table below
A full fear-killer pack (verbatim)	that week's block — packs carry per-question offsets
Chapter breakdown (module)	the week that bundles it (mapping below)
Books / YT playlists / MOOCs / practice banks	Resources section in this file
Exam-day order	Exam stacks section
Which official module covers what	Syllabus M1–M8 section

Course facts

- **Code:** MT-335 · **Credits:** 3+0 (theory only, no lab) · **Contact:** 3 lectures/week
- **Weightage:** Midterm 35% / Final 45% / Quizzes+Assignments 20% (web-verified NED policy)
- **Difficulty:** 3/5 · **GPA Risk:** Low
- **Rotation:** Course B (Mon+Tue) — 2 B-slots/wk, 0 A-slots; PS has no lab/viva
- **Midterm:** Week 8 (05–11 Oct) · **Final:** Week 15 (23–29 Nov)
- **Exam character:** 60% numerical + 40% theory, formula-driven, highly predictable. **Hypothesis testing is the single most important topic — one 15-mark question guaranteed.** Speed is the A vs A+ differentiator.

Master weekwise line-offset table

Offsets are line numbers into the named source file. `Fear-Killer-Packs.md` and `Week-by-Week-Narrative.md` are the two anchored sources; `weeks/PS-Wn.md` files live in `weeks/`.

Week	Narrative	Pack heading	Pack questions	weeks/PS-Wn	Chapter
W1	L17–32	L11 (Week 1: descriptive)	res L12 · Q1 L14 · Q2 L15 · Q3 L16	PS-W1 L1–40	01 (L1–122)
W2	L34–51	L11 (Week 1: descriptive)	res L12 · Q1 L14 · Q3 L16	PS-W2	02 (L1–140)
W3	L52–68	L18 (Week 2: probability)	res L19 · Q2 L22 · Q4 L24	PS-W3	03 (L1–128)
W4	L70–88	L18 (Week 2: probability)	res L19 · Q1 L21 · Q3 L23	PS-W4	03
W5	L90–107	L26 (Week 3: random-variables)	res L27 · Q1 L29 · Q2 L30	PS-W5	04 (L1–153)
W6	L108–126	L35 (Week 4: distributions)	res L36 · Q1 L38 · Q2 L39 · Q3 L40 · Q4 L41 · Q5 L42	PS-W6	05 (L1–139)
W7	L128–145	all packs review	—	PS-W7	01–05

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W8	L147–156	— (no pack)	—	PS-W8	—
W9	L158–179	L44 (Week 5: sampling-and-clt)	res L45 · Q1 L47 · Q2 L48 · Q3 L49	PS-W9	06 (L1–135)
W10	L180–205	L51 (Week 6: hypothesis-testing)	res L52 · Q1 L54 · Q2 L55 · Q3 L56	PS-W10	07 (L1–154)
W11	L206–225	L51 (Week 6: hypothesis-testing)	res L52 · Q4 L57	PS-W11	07
W12	L226–246	L59 (Week 7: confidence-intervals)	res L60 · Q1 L62 · Q2 L63 · Q3 L64	PS-W12	07 (CI part)
W13	L247–267	L66 (Week 8: regression-and-correlation)	res L67 · Q1 L69 · Q2 L70 · Q3 L71 · Q4 L72 · Q5 L73	PS-W13	08 (L1–146)
W14	L268–295	all packs review	—	PS-W14	01–08
W15	L297–315	— (no pack)	—	(W15 note at PS-W14 L54)	—

Pack headings are numbered by syllabus topic (Week 1–8), **not** calendar weeks (per `weeks/README.md`).

Syllabus M1–M8 → where it lives

Module	Breakdown file	Weight	Covered in
M1 Introduction to Statistics	01-Introduction-to-Statistics.md	★★★★☆☆	W1
M2 Measures of Central Tendency & Dispersion	02-Measures-of-Central-Tendency-and-Dispersion.md	★★★★☆☆	W2
M3 Probability	03-Probability.md	★★★★☆☆	W3–W4
M4 Random Variables & Random Processes	04-Random-Variables-and-Random-Processes.md	★★★★☆☆	W5 (+ W14 Markov/queuing)
M5 Probability Distributions	05-Probability-Distributions.md	★★★★☆☆	W6
M6 Sampling & Sampling Distributions	06-Sampling-and-Sampling-Distributions.md	★★★★☆☆	W9
M7 Statistical Inference & Testing of Hypothesis	07-Statistical-Inference-and-Testing-of-Hypothesis.md	★★★★★★	W10–W12 (HT W10–W11, CI W12)
M8 Simple Regression & Correlation	08-Simple-Regression-and-Correlation.md	★★★★☆☆	W13

M7 (hypothesis testing) is the highest-weight module — one 15-mark question guaranteed (per `weeks/README.md` L44).

Exam probability table (editorial — study prioritization only)

Topic	Midterm	Final	Numerical	Diagram	Theory	Definition
Descriptive stats	90%	30%	40%	15%	30%	15%
Probability basics	85%	35%	40%	10%	35%	15%
Bayes theorem	80%	45%	45%	5%	30%	20%
Random variables	75%	40%	40%	10%	30%	20%
Distributions	70%	55%	40%	10%	30%	20%
CLT/Sampling	0%	60%	35%	10%	35%	20%
Hypothesis testing	0%	95%	40%	5%	35%	20%

Chi-square	0%	70%	35%	10%	35%	20%
Confidence intervals	0%	80%	45%	5%	30%	20%
Regression	0%	65%	40%	15%	30%	15%

Marks allocation

Component	Weight	Strategy
Hypothesis testing ($z/t/\chi^2$)	~25%	4-step procedure, decision flowchart
Confidence intervals	~15%	3 formulas, sample size calc
Bayes theorem	~10%	Medical testing problems
Probability distributions	~10%	Identify + compute parameters
Regression	~10%	Least squares, $a + b$ derivation
Theory	~30%	Definitions, interpretations

Exam stacks

- **Midterm (W8, 05–11 Oct):** past paper 60 min → blank page 30 min → error log 20 min. Answer numericals first, theory second. Z-table speed: 10 lookups in 2 min. Bayes: 2 complete problems. Sleep 8 h. Ledger frozen.
- **Final (W15, 23–29 Nov):** numericals first (hypothesis testing, confidence intervals, Bayes, regression, probability) → theory (definitions, interpretations, comparisons). For hypothesis testing write the 4 steps explicitly — correct structure earns 60% of marks even if the calculation is wrong. Sleep 9 h (banked from W12).

Resources

YouTube playlists

Playlist	Channel	Videos	Why
Statistics 101 Playlists	Brandon Foltz	~60+	Best structured intro; separate playlists for probability, distributions, sampling, hypothesis testing, regression
StatQuest Statistics Fundamentals	StatQuest with Josh Starmer	~100+	Clearest explanations; hypothesis testing, p-values, CIs made simple; 1M+ subs
Probability (Stat 110)	Harvard (Joe Blitzstein)	~40+	Rigorous probability theory; mathematical foundation

Textbooks

- **Jay L. Devore** — *Probability and Statistics for Engineering and the Sciences*, 9th Ed. ISBN **978-1-305-25180-9**. Perfect for engineering students; calculus-based but intuitive; covers descriptive stats, probability, distributions, sampling, hypothesis testing, regression, ANOVA.
- **Walpole, Myers, Myers, Ye** — *Probability & Statistics for Engineers & Scientists*, 9th Ed. ISBN **978-0-13-411585-6**. Classic; strong on applications with real engineering data.

Free MOOCs

Course	Source	Notes
NPTEL: Probability and Statistics	Prof. Somesh Kumar, IIT Kharagpur	12 weeks; verified live
NPTEL: Engineering Statistics	IIT Bombay	12 weeks
MIT OCW: 18.05 Intro to Probability and Statistics	MIT	Full course
MIT RES.6-012: Introduction to Probability	John Tsitsiklis	Excellent theory
TU Delft: Probability & Statistics for Engineers	TU Delft	Engineering-applied, real datasets

Problem banks / practice

Resource	Link
Seeing Theory Interactive	https://seeing-theory.brown.edu
StatCalc: Formula Sheet + Calculators	https://www.statscalculators.com/
Khan Academy: AP Statistics	https://www.khanacademy.org/math/ap-statistics
Harvard Stat 110 Problem Sets	https://projects.iq.harvard.edu/stat110/

Cheat sheets / revision notes

Resource	Link
William Chen Probability Cheatsheet (PDF)	https://diverdi.colostate.edu/all_courses/probability%20cheatsheet.pdf
Harold's Statistical Distributions Cheat Sheet (PDF)	http://www.toomey.org/tutor/harolds_cheat_sheets/Harolds_Stats_Distributions_Cheat_Sheet.pdf
Comprehensive Statistics Formula Sheet (PDF)	https://www.statscalculators.com/resources/formula-sheet/statistics-formula-sheet.pdf
EduRev: Probability & Statistics Cheat Sheet	https://edurev.in/t/506121/data-science-cheatsheet-probability-statistics

Secret weapon

Seeing Theory — <https://seeing-theory.brown.edu> — interactive visual textbook (D3.js). Watch the CLT happen, adjust distributions in real time, experiment with Bayesian updating. Makes abstract formulas tangible.

Where-everything-lives index (source → embedded location)

Source file	~Lines	Embedded in
01-Course-Overview.md	65	NAV facts/weights + exam probability + marks allocation
02-Official-Syllabus.md	64	NAV (M1–M8 map + textbooks)
03-Chapter-Breakdowns/01 Intro to Statistics	122	BEFORE-MID W1 (verbatim)
03-Chapter-Breakdowns/02 Central Tendency	140	BEFORE-MID W2 (verbatim)
03-Chapter-Breakdowns/03 Probability	128	BEFORE-MID W3 (verbatim)
03-Chapter-Breakdowns/04 Random Variables	153	BEFORE-MID W5 (verbatim) + AFTER-MID W14 supplement
03-Chapter-Breakdowns/05 Distributions	139	BEFORE-MID W6 (verbatim)
03-Chapter-Breakdowns/06 Sampling & CLT	135	AFTER-MID W9 (verbatim)
03-Chapter-Breakdowns/07 Inference & HT	154	AFTER-MID W10 (verbatim)
03-Chapter-Breakdowns/08 Regression	146	AFTER-MID W13 (verbatim)
Definition-Book.md	25	per-week "Definitions" blocks (verbatim + L-numbers)
Formula-Book.md	29	per-week "Formulas" blocks (verbatim + L-numbers)
Diagram-Book.md	15	per-week "Diagrams" numbered refs (#1–#10)
Numerical-Book.md	42	per-week "Numericals" numbered types (#46–70)
Marks-Allocation.md	13	NAV (this file)
Resources.md	48	NAV (this file) + per-week book/video lines
Fear-Killer-Packs.md	73	per-week packs (verbatim, per-question offsets)
Week-by-Week-Narrative.md	324	per-week narrative (verbatim)
weeks/PS-W1...W14 + README	~650	per-week anchors, P0 floor, drill targets

Schedule at a glance

W	Dates	Variant	File	Topics	Chapter
1	17–23 Aug	P2	BEFORE	Descriptive statistics	01
2	24–30 Aug	P2	BEFORE	Central tendency & dispersion	02
3	31 Aug–06 Sep	P2	BEFORE	Probability theory	03
4	07–13 Sep	P2	BEFORE	Conditional probability & Bayes	03
5	14–20 Sep	P2	BEFORE	Random variables & random processes	04
6	21–27 Sep	P2	BEFORE	Probability distributions	05
7	28 Sep–04 Oct	P1	BEFORE	Midterm revision	01–05
8	05–11 Oct	MIDTERM	BEFORE	Exam week — no new material	—
9	12–18 Oct	P1	AFTER	Sampling distributions & CLT (recovery)	06
10	19–25 Oct	P0	AFTER	Hypothesis testing I: z & t	07
11	26 Oct–01 Nov	P0	AFTER	Hypothesis testing II: chi-square & Type I/II	07
12	02–08 Nov	P0	AFTER	Confidence intervals	07
13	09–15 Nov	P1	AFTER	Regression & correlation	08
14	16–22 Nov	P0	AFTER	Final taper — past paper FIRST	01–08
15	23–29 Nov	FINAL	AFTER	Exam — execute	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week.